Family Name Wilson

First Name Conrad

Student Number 29380692

Venue C1

Seat Number L24

UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

No electronic/communication devices are permitted.

No assessment materials may be removed from the assessment room.

Mathematics and Statistics

TEST

Mid-course Test

EMTH119-25SU2 (C) Engineering Mathematics 1B

Time Allowed: 2 HOURS

Assessment Conditions:

Restricted book: students may bring in the listed approved materials only.

Calculators with a 'UC' sticker permitted.

No assessment materials may be removed from the assessment room.

Materials Permitted in the Assessment Venue:

One A4 double-sided page of notes.

Materials to be Supplied to Students:

1 x Standard Write-on test paper.

1 x Formula Sheet.

Instructions to Students:

Show full working.

Marks will be lost for poorly presented or incomplete answers.

A formula sheet is provided with this test booklet.

Unless otherwise stated, give all answers in exact form.

Use blue or black pen. Do NOT use red pen.

Pencil may be used for graphs.

For Examiner Use Only

Question	Mark
Q1 /10	
Q2 /13	
Q3 /12	
Q4 /8	
Q5 /9	
Q6 /10	
Total /62	

Questions start on Page 3

$1+x$

Question 1 (10 marks)

(a) Consider the following differential equation.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1+x}{2y}, \quad y > 0$$

(4 marks)

(i) Is the differential equation linear or non-linear? Write your answer in the box.

non-linear

(ii) Use the method of separation of variables to find the general solution to the differential equation. Show full working.

Write your final answer in the box using $y(x) = \ldots$ form.

$\frac{y'}{x} - \frac{1}{2y} = \frac{1}{2y}$

$y' = f(x)g(y)$

$\int \frac{1}{g(y)}\,dy = \int fx\,dx$

$e^{\int 2}$ e^{2x}

$(1+x) \times (2y)^{-1}$

$y' 2y = 1 + x$

$\cancel{u = e^{2x}}$
$v = 1+x$
$u' = \cancel{2e^{2x}}$
$\cancel{v' =}$

$\int \frac{1}{2y^{-1}}\,dy = \int (1+x)$

$y e^{2x} = \int (1+x) e^{2x}$
$\int e^{2x} + xe^{2x}$
$2e^{2x} +$

$\int 2y\,dy = \int x + 1$

$\sqrt{y^2} = \sqrt{x^2/2 + x + C}$

$\int u\,dv = uv - \int v\,du$

$y = \sqrt{x^2/2 + x + C}$

$u = 1 + x$
$v = 2e^{2x}$

$y(x) =$

(b) Consider the following initial value problem.

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = e^{-x}, \quad y(0) = 4$$

Use the method of integrating factors to solve this initial value problem.

Write your final answer in the box using $y(x) = \ldots$ form. (3 marks)

$y' + 2y = e^{-x}$ $\quad P(x) = 2 \quad I(x) = e^{\int 2} \quad e^{2x}$

$y e^{2x} = e^{-x} e^{2x}$

$y e^{2x} = \int e^{x}$

$\frac{y e^{2x}}{e^{2x}} = \frac{e^{x} + C}{e^{2x}}$

$y = e^{-x} + C e^{-2x}$

$4 = 1 + C$

$-1 \quad -1$

$3 = C$

$y(x) = e^{-x} + 3e^{-2x}$

$\boxed{y(x) = e^{-x} + 3e^{-2x}}$

(c) A colony of *Mycobacterium tuberculosis* bacteria initially grows exponentially at a rate of 5% per hour.

(3 marks)

(i) Tick inside the circle beside the differential equation below that gives a correct model for the growth of the colony, where y is the number of bacteria at time t.

- ○ $\dfrac{dy}{dt} = 5y$
- ○ $\dfrac{dy}{dt} = 0.05 + y$
- ✓ $\dfrac{dy}{dt} = 0.05y$
- ○ $\dfrac{dy}{dt} = \dfrac{y}{5}$

(ii) Write down the solution to the differential equation if the initial number of bacteria is $y_0 = 100$.

Write your final answer in the box using $y(t) = \ldots$ form.

$y(t) = 0.05y + t$

(iii) How long does it take the number of bacteria to double? Give your answer to the nearest hour. Write your final answer in the box.

$100 \xrightarrow{5} 105 \xrightarrow{5.825} 112.5 \xrightarrow{} 118.125 \xrightarrow{5.9} 124$ $y' = 0.05y + 100$

$t(0)$ $t(1)$ $t(2)$ $t(3)$ $t(4)$

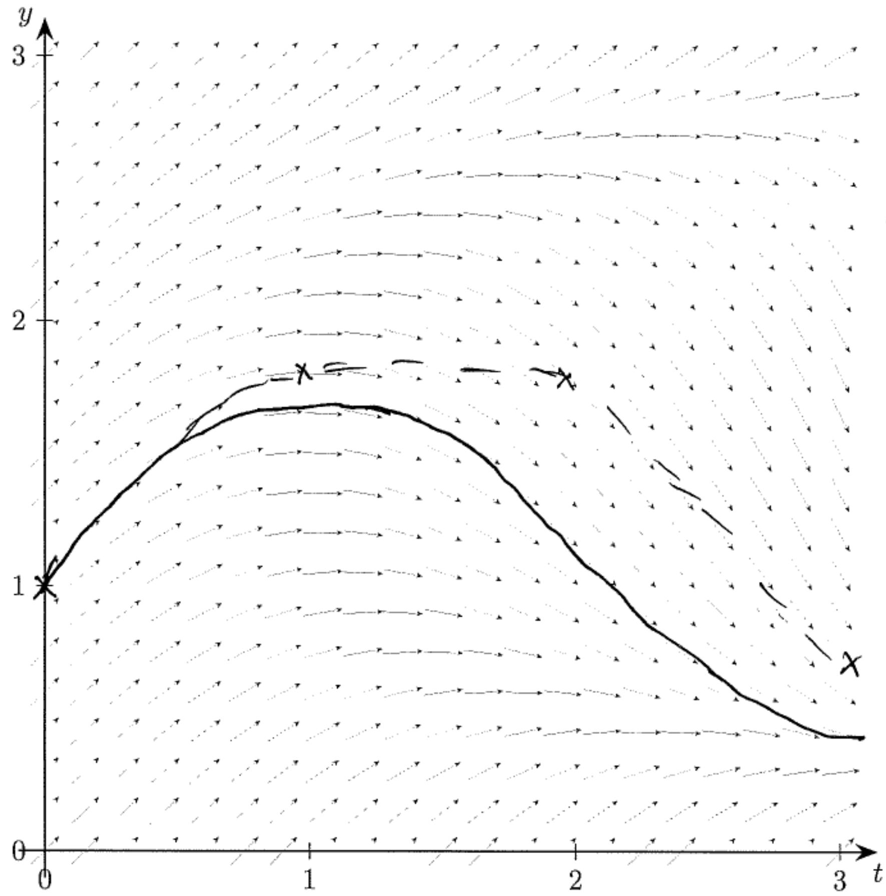
14

Question 2 (13 marks)

(a) The diagram below is a direction field for the first-order differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 1 - t\sin(y).$$

(2 marks)



(i) On the direction field, sketch the approximate solution curve with the initial data $y(0) = 1$.

(ii) Using the direction field, determine whether Euler's method with step size $h = 1$ will overestimate or underestimate the true solution for $y(3)$.

Tick inside the circle beside the correct answer.

$y_{n+1} = y_n + hf(1 - t\sin(y))$

✓ Overestimate

○ Underestimate

Your answer should be supported by a sketch on the direction field.

(b) You are given an initial value problem for a first order differential equation with $y(0) = 1$.

Write in the box the step-size you would use to estimate $y(2)$ using 10 steps of Euler's method.

(1 mark)

$h = \frac{b-a}{n} \quad h = \frac{2-0}{10} \rightarrow \frac{1}{5}$

Step-size = $1/5$

(c) Consider the following initial value problem.

$$\frac{\mathrm{d}y}{\mathrm{d}t} = y - 4t^2 + 1, \qquad y(0) = 1$$

Use two steps of Euler's method with a step-size of $1/2$ to estimate $y(1)$. Write your final answer in the box.

(2 marks)

$y_{n+1} = y + h(y - 4t^2 + 1)$

$y_1 = 1 + \frac{1}{2}(1 - 4(0)^2 + 1) = 1 + \frac{1}{2}(2) = 2$

$y_2 = 2 + \frac{1}{2}(2 - \underbrace{4(\frac{1}{2})^2}_{1/4} + 1) = (2 - 1 + 1)\frac{1}{2} + 2 = 3$

$\therefore y(1) = 3$

$y(1) \approx 3$

(d) The following second order differential equation models a spring-mass system.

$$y'' + 8y' + 16y = 0$$

(3 marks)

(i) Find the general solution of the differential equation.

Write your final answer in the box using $y(t) = \ldots$ form.

$x^2 + 8x + 16 = 0$

$(x+4)^2$

$(C_1 + C_2 x)e^{-4x}$

$y(t) = (C_1 + C_2 x$

(ii) Tick inside the circle beside the correct statement for the behaviour of this spring-mass system.

- ○ The system is over-damped.
- ○ The system is under-damped.
- ○ The system shows simple harmonic motion.
- ✓ The system shows critical damping.

(e) Consider the following initial value problem (IVP).

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 9y = 0, \quad y(0) = 2, \quad y'(0) = 0$$

(5 marks)

(i) Find the particular solution to the IVP.

Write your final answer in the box using $y(t) = \ldots$ form.

$y'' + \cancel{y'} = 0 \qquad y(0) = 2 \quad y'(0) = 0$

$y'' + 0 + 9 = 0$

$x^2 + 9 = 0$

$y(t) =$

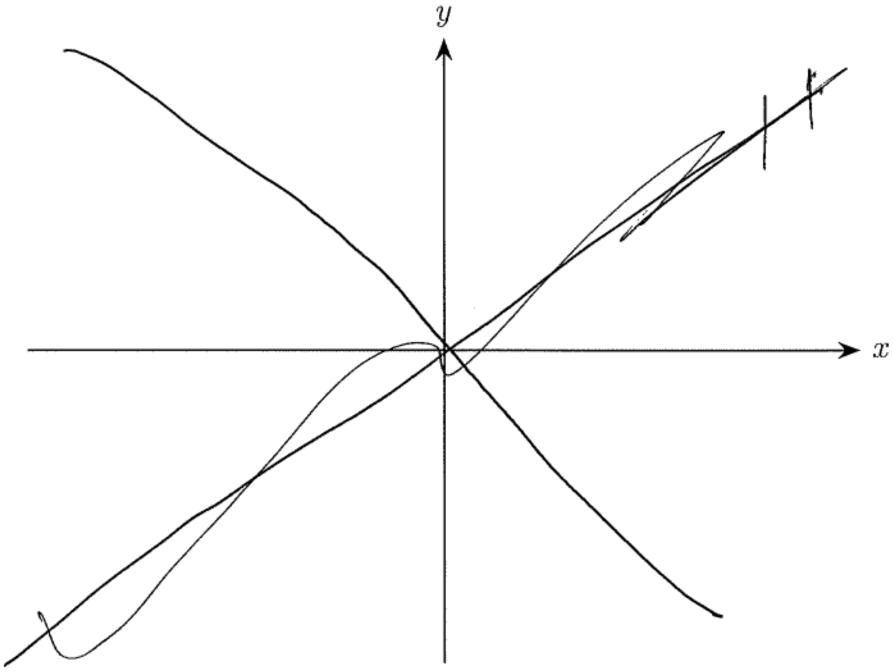
(ii) Describe in detail how the spring-mass system represented by this IVP behaves.

Question 3 (12 marks)

(a) We can use Newton's method to solve intersection problems such as $\ln(x) = -x$.

(5 marks)

(i) Using the axes below, sketch and label the functions $y = \ln(x)$ and $y = -x$. Mark any x-intercepts and y-intercepts and indicate the point of intersection.



(ii) Write the function $f(x)$ you would use in the Newton's method formula to solve this problem. Write your answer in the box.

$$f(x) = x_n - \frac{\ln(x)}{-1}$$

(iii) Use Newton's method with a suitable initial value to compute the value of x_1. Give your answer exactly or rounded to 3 decimal places. Write your final answer in the box.

$$x_{n}+1 = x_n - \frac{f(x_n)}{f'(x_n)}$$

$x_1 =$

(b) Consider the following recursively defined sequence.

$$a_{n+1} = \frac{5 + a_n}{2}, \quad a_1 = 1$$

$$\frac{5 + a_0}{2} = 1 \qquad 5 + a_0 = 2, \quad a_0 = 2 - 5, \quad a_0 = -3$$

(3 marks)

(i) Given that $a_1 = 1$, find the next three terms of the sequence, a_2, a_3, and a_4. Write your final answers in the boxes.

$$a_2 = \cancel{1/2+}\ \frac{5+1}{2} = 3$$

$$a_3 = \frac{5+3}{2} = 4$$

$$a_4 = \frac{5+4}{2} = 9/2$$

$$a_5 = \frac{9.5}{2} =$$

$a_2 = 3$

$a_3 = 4$

$a_4 = 9/2$

(ii) Given that the sequence converges, use the recursion equation to find the limit L.

Show full working and write your final answer in the box.

as $a_n \to \infty$

$L = 5$

(c) Consider the following sequence.

$$\{\cos(n\pi)\}_{n=0}^{\infty}$$

Determine whether the sequence converges or diverges, giving reasons. (1 mark)

Converges on 1 and -1 as $\cos(x) \nrightarrow 1?$ or $1<$

(d) Consider the power series $\sum_{k=0}^{\infty}(x-2)^k$. (3 marks)

(i) Find the values of x for which the series converges. Write your final answer in the box using interval notation.

$\sum_{k=0}^{\infty}(x-2)^k \rightarrow$ $|r| < 1$ converges on $\frac{a}{1-r}$

$\frac{1}{1-(x-2)}$

$$\boxed{2}$$

(ii) Find the sum of the series for the values of x for which it converges. Simplify your answer and write your final answer in the box.

$$\boxed{\sum_{k=0}^{\infty}(x-2)^k = \quad 2}$$

Question 4 (8 marks)

(a) Use the Maclaurin polynomial formula to find and simplify the order 4 Maclaurin polynomial $p_4(x)$ for the function $f(x) = \ln(1+x)$. You must show full working for marks.

(3 marks)

$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$

$\ln(1) + \frac{1}{x+1} +$

$f(x) = \ln(1+\cancel{x})$

$f'(\cancel{x} = 1/(x+1)$

$f''(x) =$

$\frac{(-1)^5 x^5}{5}$

$p_4(x) = \frac{(-1)^5 x^5}{5}$

(b) Use the pattern you see in (a) to write the Maclaurin series for $f(x) = \ln(1+x)$.

Write your answer in the box **using sigma notation**.

(1 mark)

$\ln(1+x) \sim \frac{(-1)^n x^{n+1}}{n+1}$

(c) The Maclaurin series for $f(x) = \cos x$ is given below.

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}$$

(4 marks)

(i) Use this series and the uniqueness theorem to write $\cos(x^2)$ using an order 4 polynomial and big $\mathcal{O}$ notation.

(ii) Use your answer to part (i) to evaluate the following limit.

$$\lim_{x \to 0} \frac{\cos(x^2) - 1}{x^4}$$

Write your final answer in the box.

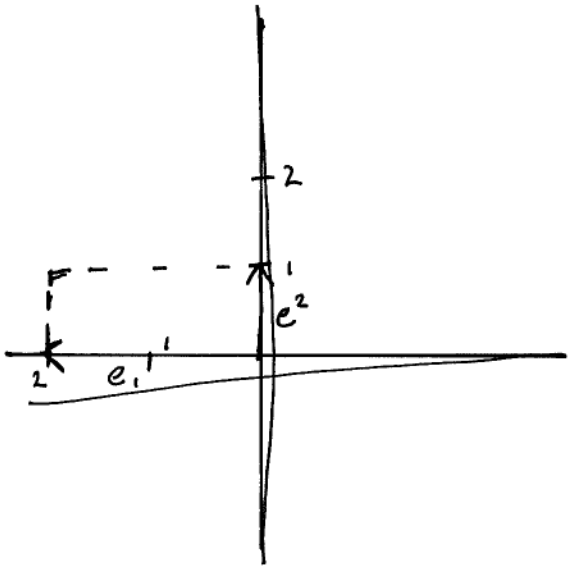
$$\lim_{x \to 0} \frac{\cos(x^2) - 1}{x^4} =$$

Question 5 (9 marks)

(a) Consider the matrix $T = \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix}$.

(5 marks)

(i) Sketch and label the image of the unit square under the transformation T.



(ii) Describe the transformations represented by the matrix T.

The unit scaled by a factor of 2 in x, and was reflected along the y axis

(iii) What is $\det(T)$?

Write your final answer in the box.

$\det(T) = -2$

(b) Use cofactor expansion to find the determinant of the following matrix. State which row or column you are using for your expansion. Write your final answer in the box.

$$M = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 2 & 1 \\ 4 & 0 & 5 \end{bmatrix}$$

(2 marks)

$\det(M) =$

(c) If A and B are 3×3 matrices with $\det(A) = 4$ and $\det(B) = -1$, find the following determinants. **Show full working.** Write your final answers in the boxes.

(2 marks)

(i) $\det(2A^{-1}B^T)$

$2^3 \times \frac{1}{4} \times -1$

$8 \times \frac{1}{4} = 2 \times -1 = -2$

$\det(2A^{-1}B^T) = -2$

(ii) $\det(A^3B^2)$

32

$\det(A^3B^2) = 32$

Question 6 (10 marks)

(a) Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$. (4 marks)

(i) Show that $\mathbf{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ is an eigenvector of A and find the corresponding eigenvalue. Write your final answer for the eigenvalue in the box.

$$\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \end{bmatrix} \begin{matrix} -1+2 \\ -5+4 \end{matrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$\boxed{-1}$

(ii) Given that $\lambda = 6$ is the other eigenvalue of A, find a corresponding eigenvector. Write your final answer for the eigenvector in the box.

$$\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} \begin{matrix} = \\ = \end{matrix}$$

(b) Consider the matrix $B = \begin{bmatrix} k-1 & 1 \\ 2 & k-2 \end{bmatrix}$.

(3 marks)

(i) For which values of k is the matrix B not invertible? Show your reasoning and write your final answers in the box.

if det = 0

$(k-1 \times k-2) - 2 = 0$

$k(-1 \times -2)$

$k(-3) - 2 = 0$

$k(-3) = 2$

$k = -\frac{2}{3}$

$k(2) = \frac{2}{2}$

$k = 1$

$1\frac{2}{3} \times 2\frac{2}{3}$

1.6×2.6

$\frac{1}{3} \times \frac{4}{3}$

$\frac{9}{9}$

$\boxed{2}$

(ii) If $k = 4$, find the characteristic polynomial and hence the eigenvalues of B. Write your eigenvalues in the box.

$\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$

(c) Consider the following matrix.

$$C = \begin{bmatrix} 3 & 1 & -2 \\ 5 & -1 & 6 \\ 0 & 0 & 4 \end{bmatrix}$$

(3 marks)

(i) Find all the eigenvalues of the matrix C. Write your final answers in the box.

(ii) Write down the eigenvalues of the matrix C^3 in the box.

End of Test

Back cover of Test.

Blank page for extra working if needed. Please write the question number if used.